Clinical Protocol Design & Feasibility

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 06 of 8 · draft protocol sections grounded in guidance and RWD — clinical leads own the design

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM REWRITES TO RIGHT-FIRST-TIME

Clinical Protocol Design & Feasibility

A governed protocol copilot that drafts sections grounded in current guidance and links feasibility assumptions to real-world data — while clinical and medical leads own the design, reducing the avoidable amendments that delay every trial.

~57%

of protocols incur ≥1 substantial amendment; ~45% avoidable

[industry-research — Tufts]

$535K

median direct cost of a substantial Phase III amendment

[industry-research — Tufts]

~44%

rise in distinct Phase II/III procedures since 2009 (complexity)

[industry-research — Tufts]

The Issue & The Cost of Doing Nothing

THE ISSUE

- First-draft protocols incorporate historical guidance and study data late, driving rework.
- Feasibility assumptions are often not linked to real-world data, so they prove optimistic.
- Phase III protocols with ≥1 substantial amendment have risen toward ~82% in recent benchmarking.
- Each substantial amendment adds direct cost and a 60–80-day cycle delay valued at ~$800K/day.

THE COST OF DOING NOTHING

~$535K / amendment

Modeled: preventing one avoidable Phase III amendment ≈ $535K direct (Tufts) + a ~60–80-day delay valued at ~$800K/day lost sales. ~45% of substantial amendments are avoidable.

- Compounding delay cost from each amendment cycle.
- Enrollment and site disruption when protocols change mid-study.

[modeled — Tufts 2016 amendment cost + Tufts 2024 delay-day]

The design line: the agent drafts sections and surfaces feasibility evidence; clinical and medical leads own the protocol design — nothing is finalized unedited.

How We Solve It — A Governed Pipeline

Retrieve current guidance and historical/RWD evidence through a scoped gateway, draft grounded sections and feasibility — then route the design to clinical leads.

1
Retrieve guidance + data
RIM guidance + historical/RWD via gateway

→

2
Draft sections
objectives, endpoints, eligibility — grounded

→

3
Link feasibility
eligibility vs. RWD cohort feasibility

→

4
HUMAN GATE
clinical / medical leads own the design

→

5
Append-only audit
draft + source lineage logged

Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

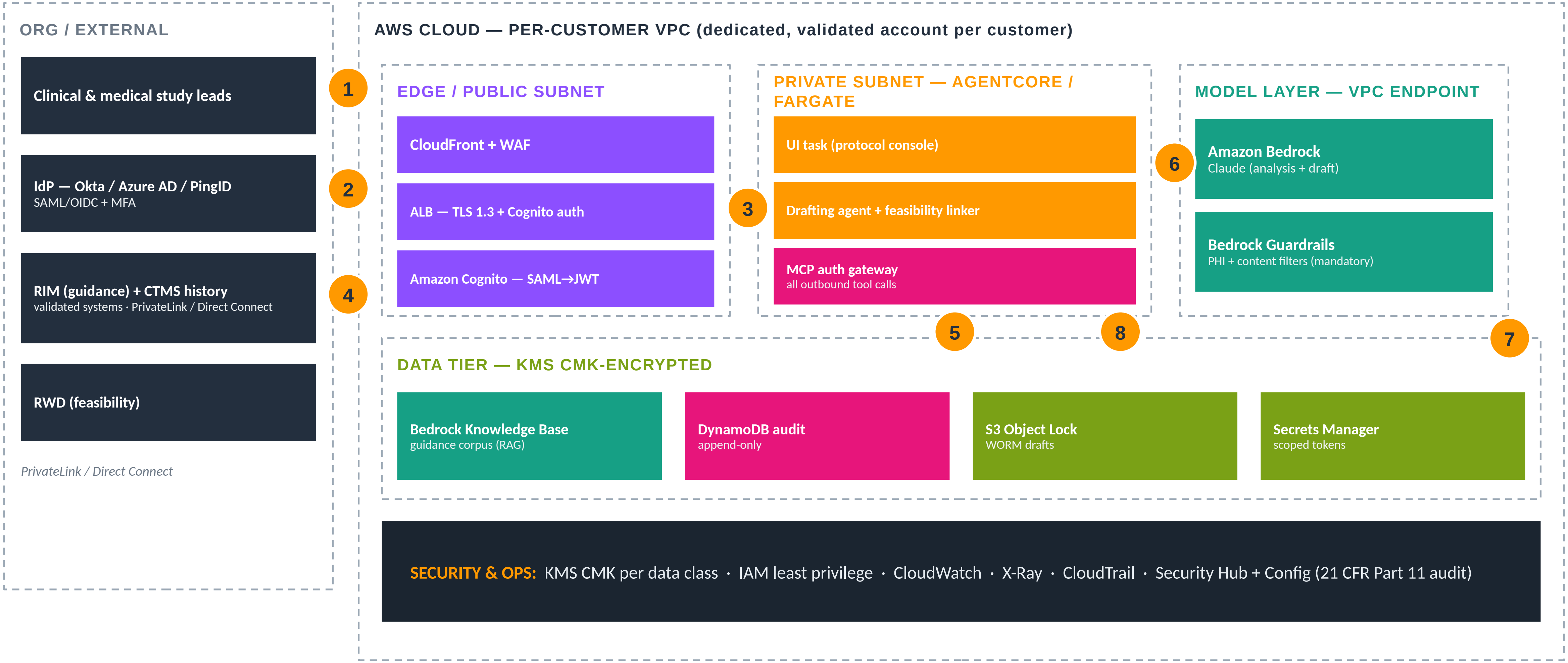
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides the protocol design

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 user sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to protocol console 4 guidance + historical/RWD reads over private connectivity 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails 7 every draft and source persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

~45%

of substantial amendments are avoidable — the target

[industry-research — Tufts]

$535K

direct cost avoided per prevented Phase III amendment

[industry-research — Tufts]

grounded

current guidance incorporated at first draft

[design property]

RWD-linked

feasibility tied to real cohort sizing

[design property]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (clinical / medical-lead roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect RIM guidance + CTMS history; connect an RWD source for feasibility.
1. Enable S3 Object Lock + append-only audit; run smoke + HITL test.

Deploy: *CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; index guidance into a Knowledge Base; connect historical + RWD sources.*

aws-native-reference/06-protocol-design/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.